

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

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Chemistry

- 1 Four metals are labelled Q, R, S, and T. The following observations are made regarding their chemical properties and extraction methods:

1. Q and R are extracted by the electrolysis of their molten compounds, whereas S and T are extracted by heating their oxides with carbon.
2. S reacts with dilute hydrochloric acid to release a gas, but T does not react.
3. When metal Q is added to an aqueous solution of a salt of R, no reaction occurs.

Which of the following shows the metals arranged in order of decreasing reactivity (most reactive first)?

- A** $Q > R > S > T$
B $R > Q > S > T$
C $R > Q > T > S$
D $S > T > R > Q$
E $T > S > Q > R$

- 2 Crude oil is separated into fractions using a fractionating column. The boiling point ranges for three of these fractions are given.

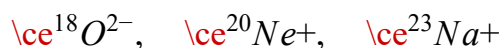
Fraction	Boiling point range / °C
Petrol	40 – 200
Kerosene	180 – 250
Diesel	250 – 350

Based on this information, which of the following statements is/are correct?

1. Diesel is collected at a higher temperature and lower down the fractionating column than petrol.
2. The intermolecular forces between molecules in the kerosene fraction are, on average, stronger than those in the petrol fraction.
3. The average number of carbon atoms per molecule is smaller in the diesel fraction than in the petrol fraction.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

3 Consider the following three gaseous ions:



Which of the following statements is/are correct?

1 All three ions have the electron configuration 2,8

2 $^{18}\text{O}^{2-}$ and $^{20}\text{Ne}^+$ contain the same number of neutrons

3 The neutron-to-proton ratio in $^{18}\text{O}^{2-}$ is greater than that in $^{23}\text{Na}^+$

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

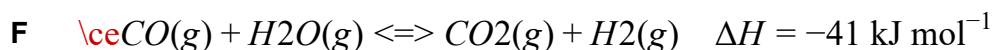
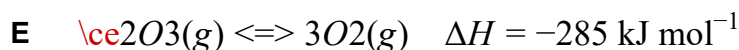
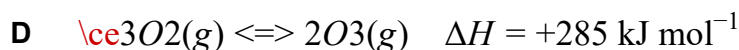
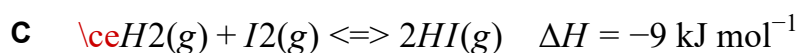
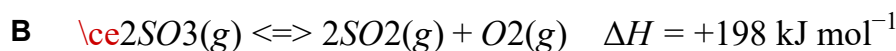
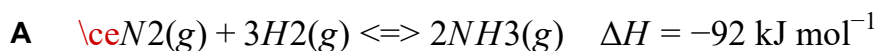
G 2 and 3 only

H 1, 2 and 3

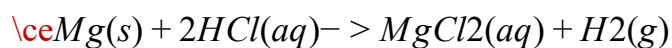
4 A reversible reaction producing a gaseous product was investigated in a closed vessel. The table below shows the percentage yield of the product at equilibrium and the time taken to reach equilibrium under different experimental conditions:

Experiment	Temperature / K	Pressure / kPa	Catalyst	Equilibrium yield / %	Time to reach equilibrium / min
1	400	100	Absent	35	45
2	500	100	Absent	18	15
3	400	200	Absent	52	20
4	400	100	Present	35	12

Which of the following equations could represent this reaction?



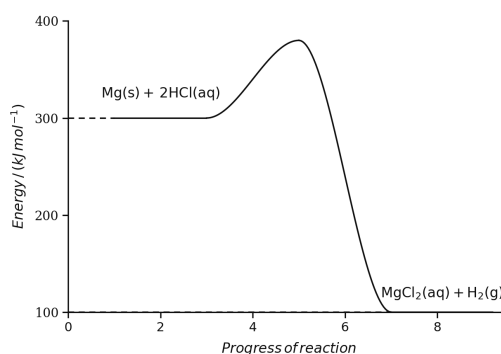
- 5 The reaction profile for the reaction between magnesium and excess dilute hydrochloric acid is shown below:



A sample of magnesium is added to an insulated container containing 150 g of dilute hydrochloric acid. Only 75% of the energy released by the reaction is absorbed by the solution, with the remainder transferred to the surroundings.

What mass of magnesium is required to increase the temperature of the solution from 18.0°C to 43.0°C?

(Assume that the specific heat capacity of the solution is 4.0 J g⁻¹°C⁻¹ and $A_r(\text{Mg}) = 24.0$.)



- A 1.35 g
- B 1.80 g
- C 2.40 g
- D 3.20 g
- E 4.80 g
- F 6.00 g

- 6 An element M forms an ionic oxide with the formula M_2O_3 .

Which row in the table gives the correct chemical formulae for the sulfate, phosphate, and nitride of M?

	Sulfate of M	Phosphate of M	Nitride of M
A	MSO_4	$M_3(PO_4)_2$	MN
B	MSO_4	MPO_4	$M(NO_3)_3$
C	$M(SO_4)_3$	MPO_4	MN
D	$M(SO_4)_3$	$M_3(PO_4)_3$	M_3N
E	$M_2(SO_4)_3$	MPO_4	MN
F	$M_2(SO_4)_3$	MPO_4	M_3N
G	$M_2(SO_4)_3$	$M_3(PO_4)_2$	MN
H	$M_2(SO_4)_3$	$M_3(PO_4)_2$	$M(NO_3)_3$

- 7 Three aqueous solutions at 25 °C are mixed together:

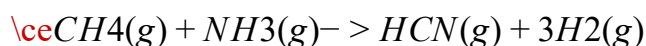
- 100 cm³ of 0.050 mol dm⁻³ $H_2SO_4(aq)$
- 150 cm³ of 0.100 mol dm⁻³ $HNO_3(aq)$
- 250 cm³ of 0.040 mol dm⁻³ $Ba(OH)_2(aq)$

Assume that H_2SO_4 , HNO_3 , and $Ba(OH)_2$ are completely dissociated in solution, and that volumes are additive.

What is the pH of the resulting mixture at 25 °C?

- A 1.0
- B 2.0
- C 3.0
- D 7.0
- E 11.0
- F 12.0
- G 13.0

- 8 The gas-phase reaction between methane and ammonia produces hydrogen cyanide and hydrogen according to the following equation:



The enthalpy change for this reaction is $\Delta H = +255 \text{ kJ mol}^{-1}$.

Bond	Mean bond enthalpy / kJ mol^{-1}
C-H	415
N-H	390
H-H	435

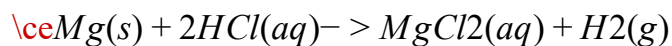
Using the data provided, what is the calculated bond enthalpy of the $\text{C} \equiv \text{N}$ triple bond in HCN ?

- A 600 kJ mol^{-1}
 B 855 kJ mol^{-1}
 C 1110 kJ mol^{-1}
 D 1270 kJ mol^{-1}
 E 1365 kJ mol^{-1}
 F 1725 kJ mol^{-1}
- 9 An aqueous solution contains an equimolar mixture of CuSO_4 , NaCl , and KNO_3 .
 The solution is electrolysed using inert graphite electrodes.

Which row in the table correctly shows the initial half-equation occurring at the cathode (negative electrode) and at the anode (positive electrode)?

	Cathode half-equation	Anode half-equation
A	$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
B	$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	$4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
C	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
D	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	$4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
E	$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
F	$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	$4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
G	$\text{K}^+(\text{aq}) + \text{e}^- \rightarrow \text{K}(\text{s})$	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
H	$\text{K}^+(\text{aq}) + \text{e}^- \rightarrow \text{K}(\text{s})$	$4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$

- 10 Magnesium reacts with dilute hydrochloric acid according to the following equation:

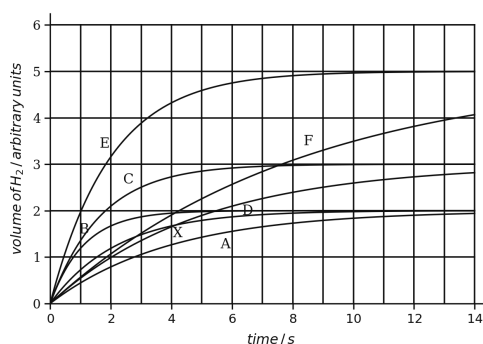


In Experiment 1, 0.48 g of magnesium ribbon was added to 100 cm³ of 0.20 mol dm⁻³ $\text{HCl}(aq)$. Curve **X** on the graph shows how the volume of hydrogen gas produced changed with time at a constant temperature.

In Experiment 2, 0.36 g of identical magnesium ribbon was added to 50 cm³ of 1.0 mol dm⁻³ $\text{HCl}(aq)$ at the same temperature and pressure.

Which curve (**A–F**) best represents how the volume of hydrogen gas produced changes with time in Experiment 2?

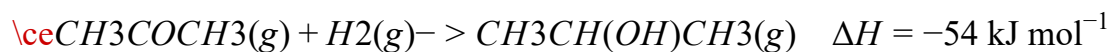
(Relative atomic mass, A_r : $\text{Mg} = 24.0$)



- A** A
- B** B
- C** C
- D** D
- E** E
- F** F

11 The structure of propan-2-ol is shown below.

Propan-2-ol can be synthesised by the gas-phase hydrogenation of propanone according to the following equation:



Mean bond energies:

$$\text{C} - \text{C} = +347 \text{ kJ mol}^{-1}$$

$$\text{C} - \text{H} = +413 \text{ kJ mol}^{-1}$$

$$\text{C} - \text{O} = +358 \text{ kJ mol}^{-1}$$

$$\text{O} - \text{H} = +464 \text{ kJ mol}^{-1}$$

$$\text{H} - \text{H} = +436 \text{ kJ mol}^{-1}$$

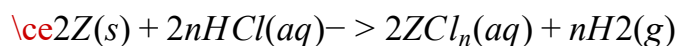
What is the mean bond energy of the $\text{C} = \text{O}$ bond in propanone?



- A 281 kJ mol⁻¹
- B 332 kJ mol⁻¹
- C 745 kJ mol⁻¹
- D 853 kJ mol⁻¹
- E 1181 kJ mol⁻¹

- 12 The graph shows the first eight successive ionisation energies of a Period 3 element, X .

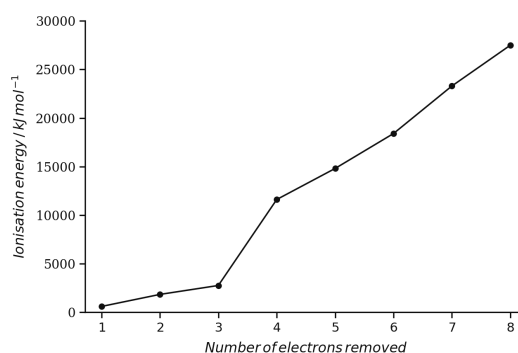
A 0.540 g sample of X reacts completely with excess dilute hydrochloric acid according to the equation:



where n is the number of valence electrons in an atom of X .

Given that the relative atomic mass of X is 27.0, what volume of hydrogen gas is produced at room temperature and pressure (r.t.p.)?

(Assume that 1.00 mol of any gas occupies 24.0 dm³ at r.t.p.)

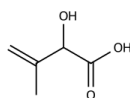


- A 240 cm³
- B 320 cm³
- C 480 cm³
- D 720 cm³
- E 960 cm³
- F 1440 cm³

13 The skeletal structure of an organic compound, X, is shown below:

Which of the following statements about compound X is/are correct?

1. 1 mol of compound X reacts with an excess of sodium metal to produce 1 mol of $\text{H}_2(\text{g})$.
2. 1 mol of compound X reacts completely with 2 mol of aqueous sodium hydroxide (NaOH).
3. Compound X exhibits both optical isomerism and *E/Z* (geometric) isomerism.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 14** Two-dimensional thin-layer chromatography (2D-TLC) is used to separate a mixture of two amino acids, X and Y.

A sample of the mixture is spotted onto the origin near one corner of a square TLC plate.

1. The plate is placed in Solvent 1. The solvent front travels 12.0 cm vertically from the baseline. The plate is then removed and dried.
2. The plate is rotated by 90° and placed in Solvent 2. The new solvent front travels 10.0 cm perpendicular to the first direction.

The retention factors (R_f) for the two amino acids in each solvent are shown in the table below:

Amino acid	R_f in Solvent 1	R_f in Solvent 2
X	0.65	0.20
Y	0.40	0.60

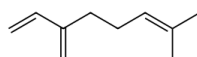
What is the straight-line distance between the centres of the spots for amino acid X and amino acid Y after the second run?

- A** 1.0 cm
- B** 3.0 cm
- C** 4.0 cm
- D** 5.0 cm
- E** 7.0 cm
- F** 10.0 cm
- G** 15.6 cm

- 15 The skeletal formula of the terpene myrcene is shown below:

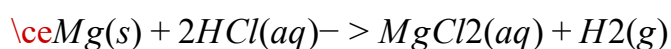
What is the minimum volume of hydrogen gas, measured at room temperature and pressure (rtp), required to completely hydrogenate 0.34 cm^3 of liquid myrcene to form the fully saturated alkane 2,6-dimethyloctane?

(M_r value: myrcene = 136. Density of myrcene = 0.80 g cm^{-3} . Molar volume of a gas at rtp = $24.0 \text{ dm}^3 \text{ mol}^{-1}$)



- A 0.048 cm^3
- B 0.096 cm^3
- C 0.144 cm^3
- D 0.288 cm^3
- E 0.048 dm^3
- F 0.096 dm^3
- G 0.144 dm^3
- H 0.288 dm^3

- 16 The reaction between magnesium and hydrochloric acid was investigated under different conditions at a constant temperature:

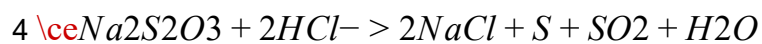
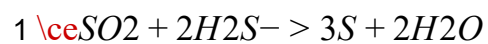


Which experiment (A–E) in the following table will produce 0.40 g of hydrogen gas in the shortest time?

(A_r values: $\text{Mg} = 24$; $\text{H} = 1.0$)

	Magnesium form	Magnesium mass / g	HCl volume / cm^3	HCl concentration / mol dm^{-3}
A	powder	2.4	250	2.0
B	powder	4.8	150	2.0
C	ribbon	4.8	100	4.0
D	powder	3.6	300	1.5
E	ribbon	4.8	200	2.0

17 Four chemical equations involving sulfur compounds are shown below:



Which two equations represent disproportionation reactions?

A 1 and 2

B 1 and 3

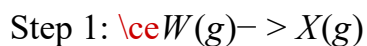
C 1 and 4

D 2 and 3

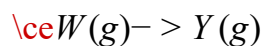
E 2 and 4

F 3 and 4

18 A gas-phase reaction proceeds via a two-step mechanism:



The overall reaction is:



The following thermodynamic and kinetic values are given:

- Enthalpy change for Step 1: $\Delta H_1 = +30 \text{ kJ mol}^{-1}$
- Activation energy for the forward reaction of Step 1: $E_{a,1} = +80 \text{ kJ mol}^{-1}$
- Overall enthalpy change for $W(g) \rightarrow Y(g)$: $\Delta H_{\text{overall}} = -40 \text{ kJ mol}^{-1}$
- Activation energy for the forward reaction of Step 2: $E_{a,2} = +50 \text{ kJ mol}^{-1}$

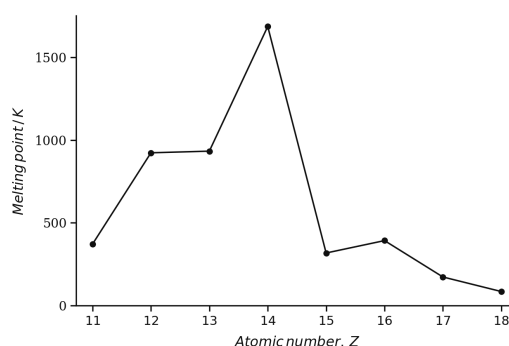
What is the enthalpy change for Step 2 (ΔH_2) and the activation energy for the reverse reaction of Step 2 ($Y(g) \rightarrow X(g)$)?

	Enthalpy change for Step 2 / kJ mol ⁻¹	Activation energy for reverse of Step 2 / kJ mol ⁻¹
A	-70	+20
B	-70	+80
C	-70	+120
D	-10	+40
E	-10	+60
F	+70	+20
G	+70	+120
H	+10	+60

- 19** The graph shows the melting points of the eight consecutive elements across Period 3 of the Periodic Table (atomic numbers $Z = 11$ to 18).

Which of the following statements correctly explain(s) features of this graph?

- 1 The sharp decrease in melting point from $Z = 14$ to $Z = 15$ occurs because the structure changes from a giant covalent network to a simple molecular structure.
- 2 The melting point of the element with $Z = 16$ is higher than that of the element with $Z = 15$ because S8 molecules have more electrons than P4 molecules, resulting in stronger London dispersion forces.
- 3 The element with $Z = 18$ has the lowest melting point because it exists as diatomic molecules with fewer electrons per molecule than Cl2.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** Aqueous solutions of two different monoprotic acids, Acid X and Acid Y, are prepared. Both solutions have a concentration of 0.10 mol dm^{-3} .

The pH of the solution of Acid X is 1.0, and the pH of the solution of Acid Y is 3.0.

Which of the following statements is correct?

- A** Equal volumes of Acid X and Acid Y require the same volume of 0.10 mol dm^{-3} NaOH solution for complete neutralisation.
- B** The solution of Acid Y has a higher electrical conductivity than the solution of Acid X.
- C** If 10 cm^3 of the solution of Acid Y is diluted to 100 cm^3 with distilled water, the pH of the resulting solution will be 4.0.
- D** The concentration of the anion X^- is approximately equal to the concentration of the anion Y^- .
- E** When reacted with an excess of magnesium ribbon, the initial rate of hydrogen gas production is greater for Acid Y than for Acid X.

- 21** A neutral atom of element X has electrons in three occupied electron shells, and its $+2$ ion has the electron configuration of neon.

Which row of the table correctly compares the elements immediately adjacent to X in the Periodic Table with element X?

	element immediately above X	element immediately below X	element immediately to the left of X	element immediately to the right of X
A	has a higher first ionisation energy than X	reacts more vigorously with cold water than X	its stable cation has a smaller ionic radius than X^{2+}	has a higher first ionisation energy than X
B	has a lower first ionisation energy than X	reacts more vigorously with cold water than X	its stable cation has a larger ionic radius than X^{2+}	has a higher first ionisation energy than X
C	has a higher first ionisation energy than X	reacts more vigorously with cold water than X	its stable cation has a larger ionic radius than X^{2+}	has a lower first ionisation energy than X
D	has a higher first ionisation energy than X	forms a hydroxide that is less soluble in water than that of X	its stable cation has a larger ionic radius than X^{2+}	has a higher first ionisation energy than X
E	has a lower first ionisation energy than X	forms a hydroxide that is less soluble in water than that of X	its stable cation has a smaller ionic radius than X^{2+}	has a lower first ionisation energy than X

- 22 The first six successive ionisation energies of a Period 3 element, Z, are shown below:

Ionisation energy	1st	2nd	3rd	4th	5th	6th
Value / kJ·mol ⁻¹	789	1577	3232	4356	16091	19785

Element Z reacts completely with excess chlorine gas to form a single liquid chloride.

Which row in the table correctly identifies the formula of this chloride, the molecular shape of this chloride, and the pH of the resulting mixture when this chloride is added to an excess of pure water?

	Formula of chloride	Molecular shape	pH of resulting mixture
A	ZCl_3	trigonal planar	<7
B	ZCl_3	pyramidal	>7
C	ZCl_4	square planar	<7
D	ZCl_4	square planar	=7
E	ZCl_4	tetrahedral	<7
F	ZCl_4	tetrahedral	=7
G	ZCl_5	trigonal bipyramidal	<7
H	ZCl_5	tetrahedral	=7

- 23** Consider the following three balanced chemical equations involving nitrogen-containing compounds:



Which of the following statements is/are correct?

1. In reaction 1, nitrogen is both oxidised and reduced.
2. In reaction 2, nitrogen undergoes disproportionation.
3. In reaction 3, hydrogen peroxide acts as a reducing agent.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 24** Consider only the three Group 2 elements Mg , Ca , and Sr , and only the three Group 16 elements O , S , and Se .

Which of the following statements is/are correct for the compound strontium oxide?

- 1 It is formed from the Group 2 element with the lowest first ionisation energy.
- 2 It is formed from the Group 16 element with the highest electronegativity and the Group 2 element with the lowest reactivity with cold water.
- 3 It is formed from the Group 16 element with the lowest boiling point.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

25 Gas-liquid chromatography was used to analyse an unknown liquid mixture.

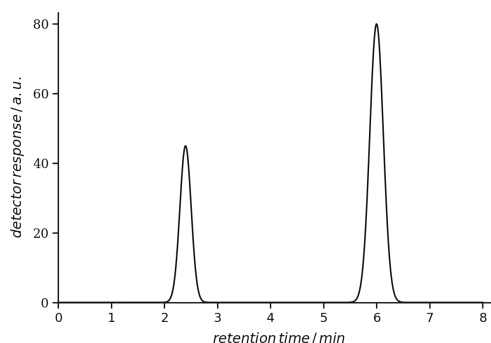
Samples of three pure reference compounds, X, Y, and Z, were each analysed separately under identical conditions (same stationary phase, column temperature, and carrier gas flow rate). Their retention times were recorded as follows:

- Compound X: 2.4 min
- Compound Y: 4.8 min
- Compound Z: 6.0 min

A sample of the unknown mixture was then injected under the exact same conditions, producing the chromatogram shown below.

Which of the following statements is/are correct?

- 1 The inert carrier gas passing through the column acts as the mobile phase.
- 2 The unknown mixture contains compounds X and Z, but does not contain compound Y.
- 3 Compound Z has a longer retention time than compound X because compound Z has a stronger affinity for the mobile phase than compound X.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 26** An oxyanion has the formula XO_4^{2-} , where X represents an isotope of an element with atomic number z and mass number $2z + 6$.

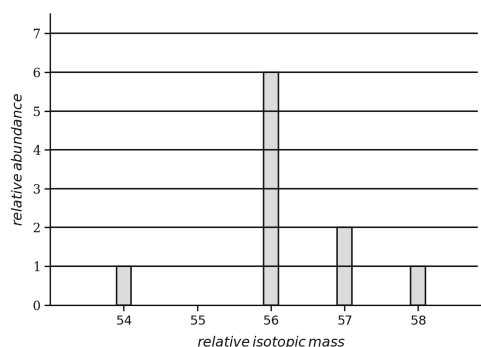
Each oxygen atom present in the ion is the isotope $^{16}_8\text{O}$.

Which row in the table gives the number of protons, neutrons, and electrons in this XO_4^{2-} ion?

	protons	neutrons	electrons
A	$z+32$	$z+38$	$z+30$
B	$z+32$	$z+38$	$z+32$
C	$z+32$	$z+38$	$z+34$
D	$z+32$	$2z+38$	$z+34$
E	$z+32$	$2z+70$	$z+34$
F	$z+34$	$z+38$	$z+34$

- 27** The mass spectrum shows the relative abundances of the isotopes of an element.

What is the relative atomic mass (A_r) of this element?



- A** 56.0
- B** 56.2
- C** 56.25
- D** 56.3
- E** 56.5